

SmartCafe Online Ordering System

Project Overview

SmartCafe is a growing coffee shop chain that wants a web-based ordering platform. Customers should be able to browse menu items, place orders online, make payments, and track order status in real-time.

The current process is completely manual and handled at the counter. Management wants to reduce waiting times and improve customer experience.

Business Goals

1. Increase online sales by 40% within six months.
 2. Reduce in-store waiting time by at least 50%.
 3. Improve customer retention through loyalty rewards.
 4. Allow staff to manage orders efficiently.
 5. Generate business analytics for management.
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User Types

Customer

People ordering food and beverages.

Store Staff

Employees preparing and managing orders.

Store Manager

Responsible for store operations and reporting.

System Administrator

Manages platform settings and user permissions.

Functional Requirements

User Authentication

- Customers can register using email and password.
- Customers can log in and log out.
- Password reset functionality should be available.
- Social login using Google should be supported.

Menu Management

- Display categories such as Coffee, Tea, Snacks, and Desserts.
- Show item name, description, image, price, and availability.
- Search menu items.
- Filter menu items by category.

Ordering System

- Add items to cart.
- Update item quantities.
- Remove items from cart.
- Place orders online.
- Support pickup orders.
- Display estimated preparation time.

Payments

- Online payment through credit/debit cards.
- Support UPI payments.
- Generate payment receipts.
- Store payment history.

Order Tracking

- Customers can track order status.

Statuses:

- Order Received
- Preparing
- Ready for Pickup
- Completed

Loyalty Program

- Customers earn reward points for purchases.
- Points can be redeemed during checkout.
- Customers can view reward balance.

Notifications

- Email confirmation after order placement.
- Order status update notifications.
- Payment confirmation notifications.

Admin Dashboard

- View daily orders.
- View revenue reports.
- Manage menu items.
- Manage customer accounts.
- View top-selling products.

Analytics

- Daily sales reports.
 - Weekly sales reports.
 - Monthly sales reports.
 - Customer purchase trends.
 - Revenue forecasting dashboard.
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Non-Functional Requirements

Performance

- System should support 10,000 daily users.
- Page load time should be under 2 seconds.
- Search results should appear within 1 second.

Security

- Passwords must be encrypted.
- HTTPS must be enforced.
- Role-based access control required.
- Secure payment processing required.

Scalability

- Platform should support multiple store locations in future.
- Architecture should support horizontal scaling.

Reliability

- 99.9% uptime target.
- Daily backups required.

Accessibility

- Mobile responsive design.
 - WCAG accessibility compliance.
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Business Rules

1. Customers can redeem a maximum of 500 reward points per order.
 2. Orders cannot be cancelled after preparation starts.
 3. Discounts cannot be combined with loyalty point redemption.
 4. Managers can edit menu prices.
 5. Staff cannot access financial reports.
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Assumptions

- Internet connectivity is available.
 - Payment gateway provider is available.
 - Email service provider is available.
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Risks

1. Payment gateway downtime.
 2. High traffic during promotional campaigns.
 3. Incorrect inventory availability.
 4. Delayed notification delivery.
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Success Metrics

- 40% increase in online sales.
 - 50% reduction in waiting time.
 - 4.5+ average customer rating.
 - Less than 1% payment failure rate.
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Future Enhancements

- Mobile application.
- AI-powered product recommendations.
- Voice ordering.
- Subscription coffee plans.
- Integration with food delivery partners.